To be clear: any vector space in which there is a defined inner product is called an <u>inner product space</u>

Now we mention some further properties all inner products must satisfy. These should all in a sense be familiar as we saw all of these with dot product, so these are just generalizations.

(1) $\| v \|^2 = \langle v, v \rangle$

Back in dot product we first defined length via sum of of squares of the vector's components. Definitely can't do that here as there may not be "components" and even if there are, might not be real numbers.

But fortunately, the other way of looking at it, as square of length equaling the dot product of vector with itself, still works, just with inner product replacing dot product.

(2) $\| v \| = 1$ ----> v is a unit vector

Use this in potential orthonormal basis just as before.

(3) Distance between u and v is $\| u - v \|$

Can use this still in projections and best approximation ideas just as we did before.

(4) u and v are orthogonal exactly when $\langle u, v \rangle = 0$

Here also can use ideas of projections as well as Gram-Schmidt, again in any inner product.

A newish idea (ok there is still counterpart in dot product, but this is much broader than it is there):

Cauchy-Schwarz Inequality

In any inner product space, for any u, v in the space:

$$|\langle u, v\rangle| \leq \|u\| \, \|v\|$$

With dot product we had:

$$u \cdot v = \|u\| \, \|v\| \cos \theta$$

which yields this inequality in that specific scenario. That is proved via Pythagoras/Law of Cosines style ideas.

How to prove Cauchy-Schwarz in general? Can't use the geometric style ideas like Law of Cosines used in dot product, the inner product space we are working in could be totally different in structure compared to

Turns out it uses ideas of orthogonality, similar to in that respect to something we did in dot product anyway.

To start, note that for v any vector and $\hat{v}$ the projection of v on any subspace we always have

$$\| \hat{v} \| \leq \| v \|$$

This is actually a very strong statement. It would be one thing to say such and such vector has length at least as large as its projection on a particular subspace But this says it's true no matter the vector and no matter which subspace you are projecting on, that's much stronger!

Why true? Showed earlier with dot product, use orthogonal decomposition:

$$v = \hat{v} + z \qquad \text{where } \hat{v}, z \text{ orthogonal}$$

---->

$$\langle v, v\rangle = \langle \hat{v} + z, \hat{v} + z\rangle$$

---->

$$\| v \|^2 = \| \hat{v} \|^2 + \| z \|^2 + 2 \langle v, z\rangle$$

---->

$$\| v \|^2 = \| \hat{v} \|^2 + \| z \|^2$$

since v, z are orthogonal. And this shows $\| \hat{v} \| \leq \| v \|$

So now consider the projection $\hat{v}$ of v on u.

That is, the projection of v on subspace generated by u.
Via the same style formula used in dot product, we have

$$\hat{v} = \frac{\langle v, u\rangle}{\langle u, u\rangle} u$$

Taking length of both sides, we then get

$$\| \hat{v} \| = \frac{|\langle v, u \rangle|}{\| u \|^2} \| u \|$$

and that becomes

$$|\langle v, u \rangle| = \| u \| \| \hat{v} \| \leq \| u \| \| v \|$$

Cauchy-Schwarz

Similar reasoning as dot product as $\| \hat{v} \| \leq \| v \|$ is same kind of thing as $v \cos \theta$ that appeared in dot product to help prove the statement there.

This helps prove another result we saw in dot product, well known idea:

Triangle Inequality

$$\| u + v \| \leq \| u \| + \| v \|$$

Why?

$$\begin{aligned} \| u + v \|^2 &= \langle u + v, u + v \rangle \\ &= \| u \|^2 + \| v \|^2 + 2\langle u, v \rangle \\ &\leq \| u \|^2 + \| v \|^2 + 2 \| u \| \| v \| \\ &= (\| u \| + \| v \|)^2 \end{aligned}$$

Cauchy-Schwarz

----> desired result

For examples we started with regular dot product. Then slight modification of it with different coefficients.

After that an inner product on $\mathbb{P}_n$ that we defined last time.

Now a Calculus based one.

<u>Ex:</u> Consider $C[a, b]$ the set of continuous functions on $[a, b]$ into $\mathbb{R}$. For any $f, g \in C[a, b]$ define

$$\langle f, g \rangle = \frac{1}{b - a} \int_a^b f(x)g(x)\, dx$$

This is average value idea.

Why is this an inner product? $\langle f, g \rangle = \langle g, f \rangle$ is clear since fg and gf are same function. Also

$$\langle cf + g, h \rangle = c \langle f, h \rangle + \langle g, h \rangle \quad \text{works since}$$

$$\int [cf + g]h = c \int fh + \int gh$$

via properties of integrals which shows that. Then

$$\langle f, f\rangle = \frac{1}{b-a}\int_a^b [f(x)]^2\,dx$$

And this can only be 0 if $f = 0$ identically since otherwise f^2 would be positive in some portion, guaranteeing the integral in question would be positive.

Hence this is an inner product. Sometimes define it without the $\frac{1}{b-a}$ factor as that doesn't impact the dynamics. Of course values might change but not idea.

6.7.3 $\quad p(t) = 4 + t \quad , \quad q(t) = 5 - 4t^2$

at points -1, 0, 1

$$\langle p, q\rangle = p(-1)q(-1) + p(0)q(0) + p(1)q(1)$$
$$= 3(1) + 4(5) + 5(1) = 28$$

6.7.27 $f(t) = 1 - 3t^2$, $g(t) = t - t^3$

on $C[0, 1]$.

$$\langle f, g\rangle = \int_0^1 (1 - 3t^2)(t - t^3)\, dt$$

$$= \int_0^1 [3t^5 - 4t^3 + t]\, dt$$

$$= \left[\frac{t^6}{2} - t^4 + \frac{t^2}{2}\right]\Bigg|_0^1 = 0$$

So these are orthogonal.

In both these examples, the two functions we took generate a subspace. What if we wanted an orthogonal basis? Could use Gram-Schmidt to find it. Again use same formula just with inner product instead of dot product.

Ok second one was already orthogonal but point remains - those ideas work in any inner product space.

What if we wanted function of type $h(t) = e^{kt}$ orthogonal to f in second example. Need to integrate fh then. How? Use integration by parts and solve.

That's one thing about this, need to start getting back into the mode of doing derivatives/integrals as we near the differential equations part of the course.

Integration by parts, substitution, and trigonometric integrals are all relevant for sure.

Section 6.7 doesn't in a sense have a lot of content, perhaps Cauchy-Schwarz is the only really new thing as inner product ideas were already familiar to us really via dot product.

But even though not a ton of content, it's a section that tends to be quite difficult for folks.

Why? Largely due to nuance.

Say you are given an operation on a vector space and asked to check if it is an inner product. Yet the vector space/operation are totally unfamiliar to us. Well that can happen, that's part of it, certainly can be difficult, but you need to be able to do that.

Or what if given an inner product you've never seen before and need to use Gram-Schmidt with it? Well with the integral based inner product mentioned recently, will require integration during process, obviously was not typical for Gram-Schmidt earlier in chapter.

Again, that's the issue, same ideas yes, but setting can vary wildly and having to adapt to that can be difficult.

Chapter 7

In Chapter 5, the main thing we focused on was diagonalizing matrices. In Chapter 6 one very common concept was orthogonality. Early part of this chapter is aimed at putting those ideas together. Part of goal of this is to aid with the culmination of linear algebra in this class, 7.4, where we break down generic matrices.

7.1 Symmetric Matrices/Diagonalization

Real $n \times n$ matrix A is <u>symmetric</u> if

$$A^T = A$$

Yet another use of transpose. Now such a matrix, as mentioned, must be square, otherwise equalling its transpose wouldn't even make sense. Being real isn't required in comparison, but certainly in this class that is the typical setting so we assume that unless stated.

Such matrices are very important in part because they have some very special properties:

(1) Two eigenvectors which correspond to different eigenvalues are always orthogonal.

Remember in a generic real square matrix, a set of real eigenvectors which corresponded to distinct real eigenvalues was linearly independent. But this is saying something much stronger, that such a set of eigenvectors is an orthogonal set (much stronger condition as orthogonal set of nonzero vectors is linearly independent anyway).

Why? Say λ_1, λ_2 are distinct real eigenvalues with corresponding eigenvectors v_1, v_2. Observe that

$$\lambda_1 v_1^T v_2 = \lambda_1 (v_1 \cdot v_2) = (\lambda_1 v_1) \cdot v_2 = (\lambda_1 v_1)^T v_2$$

So since λ_1 is an eigenvalue for v_1 this further becomes

$$\lambda_1 v_1^T v_2 = (Av_1)^T v_2$$

$$= v_1^T A^T v_2 \qquad \text{via product of transpose}$$

$$= v_1^T A v_2 \qquad \text{because } A \text{ is symmetric}$$

$$= v_1^T \lambda_2 v_2 \qquad \text{since } \lambda_2 \text{ eigenvalue for } v_2$$

$$= \lambda_2 v_1^T v_2$$

Since λ_1, λ_2 are distinct, only way this can hold is to have $v_1^T v_2 = 0$, proving that v_1, v_2 are orthogonal as desired.

Next important property is

(2) For A real, symmetric, A always has only real eigenvalues.

Note earlier when talking about eigenvectors being orthogonal, we might have wondered what that meant if eigenvalues were nonreal making for nonreal eigenvectors. Well this guarantees no issue in that respect, only talking about real eigenvalues.

Why is this true? Well we actually proved it in HW 5.5.27, 5.5.28, was shown there if $A^T = A$ (i.e A is symmetric), then for q an eigenvalue of A, $\bar{q} = q$, that is q must be real, as desired.

This helps bring us to the concept that a real, square matrix A is <u>orthogonally diagonalizable</u> if there exists a real, orthogonal matrix P and a real, diagonal matrix D for which

$$A = PDP^{-1} = PDP^{T}$$

automatic since P orthogonal

So orthogonal diagonalization is a very special type of diagonalization. Again same conditions as usual, but P must be an orthogonal matrix on top of everything else.

So when is a real, square matrix orthogonally diagonalizable? Well seems like relates to symmetric since symmetric matrix has orthogonal eigenvectors for distinct eigenvalues.

As it turns out, a matrix which is orthogonally diagonalizable is always symmetric. Why? Say A is such. Then

$$A = PDP^{T}$$

for P orthogonal and D diagonal. From this

$$A^T = (PDP^T)^T = (P^T)^T D^T P^T = PDP^T = A$$

So that comes out fairly easily. But what about other way? If A is symmetric, is A always orthogonally diagonalizable?

That's much harder. To start with if A is symmetric is it even diagonalizable (over $\mathbb{R}$)?

Even that's not immediately clear. Sure A cannot have nonreal eigenvalues, so that's one thing we don't have to worry about. But what about repeated real eigenvalues? Those sometimes prevent a matrix from being diagonalizable so we have to be wary of that.

And that is just talking about diagonalization, orthogonal diagonalization is a much stronger condition!

Even if repeated eigenvalue hypothetically doesn't prevent diagonalization in setting of symmetric matrix, once some eigenvalue(s) are repeated and thus require multiple eigenvectors corresponding to them for diagonalization, will those eigenvectors be orthogonal?

We showed two eigenvectors for distinct eigenvalues had to be orthogonal, but two linearly independent eigenvectors from same eigenvalue? Not clear!
So quite a bit needs to cleared up here.